

Exercise 2: Implementing the Factory Method Pattern

CODE:

```
// Document Interface
interface Document {
    void open();
}

// Word Document
class WordDocument implements Document {
    public void open() {
        System.out.println("Word document opened.");
    }
}

// PDF Document
class PdfDocument implements Document {
    public void open() {
        System.out.println("PDF document opened.");
    }
}

// Excel Document
class ExcelDocument implements Document {
    public void open() {
        System.out.println("Excel document opened.");
    }
}

// Abstract Factory
abstract class DocumentFactory {
    abstract Document createDocument();
}

// Word Factory
```

```
class WordFactory extends DocumentFactory {
    Document createDocument() {
        return new WordDocument();
    }
}

// PDF Factory
class PdfFactory extends DocumentFactory {
    Document createDocument() {
        return new PdfDocument();
    }
}

// Excel Factory
class ExcelFactory extends DocumentFactory {
    Document createDocument() {
        return new ExcelDocument();
    }
}

// Main Class
public class Main {
    public static void main(String[] args) {
        DocumentFactory wordFactory = new WordFactory();
        Document word = wordFactory.createDocument();
        word.open();

        DocumentFactory pdfFactory = new PdfFactory();
        Document pdf = pdfFactory.createDocument();
        pdf.open();

        DocumentFactory excelFactory = new ExcelFactory();
        Document excel = excelFactory.createDocument();
        excel.open();
    }
}
```

}

}

OUTPUT:

```
Word document opened.  
PDF document opened.  
Excel document opened.
```
